

The Role of Manganese Carbonate Precipitation in Controlling Fluoride and Uranium Mobilization in Groundwater

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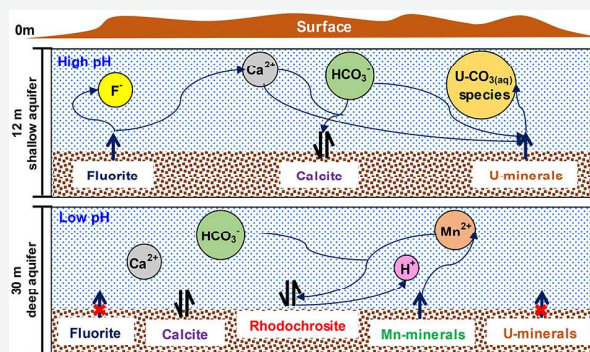
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Supporting Information

ABSTRACT: Groundwater contamination with fluoride (F) and uranium (U) has been reported in many parts of India. However, the sources and mobilization mechanisms of these contaminants remain poorly understood. The present study aimed to identify the processes governing the coexistence of elevated F and U in groundwater at a typical site in India's middle Gangetic plain. Sustained groundwater sampling at 21 locations over two years indicated persistence of high F and U in a shallow aquifer (12 m depth), but absence of these pollutants in a deeper aquifer (30 m depth). For both the aquifers, Mn exhibited strong inverse correlations with F (-0.587 ; $p < 0.01$) and U (-0.581 ; $p < 0.01$). X-ray diffraction analysis of representative sediment cores indicated few differences in the mineralogy of the two aquifers, which consisted of fluorite ($\text{CaF}_2(\text{s})$) and calcite ($\text{CaCO}_3(\text{s})$), among others. Analysis of groundwater speciation and saturation state and sequential extraction on aquifer sediments suggested that elevated F in shallow groundwater occurred due to calcite precipitation-induced fluorite dissolution. The conditions in both the aquifers were oxidizing with respect to U, but reducing with respect to Mn. Elevated U was attributed to carbonate-promoted mobilization from iron–manganese (Mn) and residual sediment fractions. In the deeper groundwater, elevated Mn and lower pH levels persisted with conditions at saturation with respect to rhodochrosite ($\text{MnCO}_3(\text{s})$). Furthermore, medium (30 d) to long-term (300 d) batch experiments were performed to systematically evaluate the role of variable Mn on calcite precipitation under approximate in situ conditions. Precipitation of rhodochrosite outcompeted calcite precipitation and resulted in lower pH compared to pH of Mn-free systems, which (a) inhibited calcite precipitation and associated fluorite dissolution and (b) constrained pH and alkalinity in the deeper groundwater. These findings have implications for understanding F and U mobilization in comparable Mn-deficient sites and development of appropriate Mn-based amendments for in situ remediation.

KEYWORDS: groundwater, aquifer sediment, fluoride, uranium, rhodochrosite, precipitation



INTRODUCTION

India is facing challenges to maintain its groundwater quality.¹ Among other pollutants, widespread contamination of groundwater with fluoride (F)^{1–3} and uranium (U)^{4–6} has influenced water availability for potable and agricultural use. Although F is an essential nutrient, which helps develop tooth enamel, excessively high F intake can cause fluorosis of tooth and bones,^{7,8} bilateral lameness, and stiffness of gait.^{9,10} F contamination in groundwater is mostly attributed to dissolution of F-bearing minerals such as fluorite [CaF_2], muscovite [$\text{KAl}_2(\text{AlSi}_3\text{O}_{10})(\text{F},\text{OH})_2$], biotite [$\text{K}(\text{Mg},\text{Fe})_3(\text{AlSi}_3\text{O}_{10})(\text{F},\text{OH})_2$], hornblende [$(\text{Ca},\text{Na})_{2,3}(\text{Mg},\text{Fe},\text{Al})_5(\text{Si},\text{Al})_8\text{O}_{22}(\text{OH},\text{F})_2$], and fluorapatite [$\text{Ca}_5(\text{PO}_4)_3\text{F}$].^{11,12} The extent of contamination depends on the types of F minerals present in the aquifers, the respective solubilities, and the levels of co-ions present.^{13,14} A widely accepted mechanism of F contamination in carbonate-rich

aquifers is calcite [CaCO_3]/dolomite [$\text{CaMg}(\text{CO}_3)_2$] precipitation-induced fluorite dissolution.^{15–17}

Uranium is another contaminant which is receiving widespread attention in recent years in India. Studies have identified critical contamination in southwest Punjab,^{18–21} Haryana, and Rajasthan.^{22,23} Uranium at elevated concentrations in groundwater is a concern for its potential chemical toxicity, rather than its radiotoxicity. Although uranium has not yet been proven to be carcinogenic,²⁴ it is known to accumulate in and cause toxicity to kidneys, liver, and bones.²⁵ The World Health Organization prescribes drinking

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